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6.10.5 Installation

6.10.5.1 General

- i. Motors stored outdoor and over a period of time before commissioning shall be provided with initial and thereafter periodic preservation procedures according to Manufacturer's instructions. As a minimum requirement, preservation shall include proper sealing of motor shaft, cable entries, turning of shafts and for those with anti-condensation heaters it should be energised.
- ii. Location of motor remote control unit (RCU) and or emergency switch should be on the opposite side of the terminal box and not be more than 1.5 metres away. Site ammeter, if installed, should be visible to operator when performing other activities, e.g. operating valve.
- iii. There shall not be any obstruction to the air flow of the motor or the motor heat exchanger.
- iv. Shim plates used should not be of non-magnetic material. Dowel pins may be installed to facilitate reinstallation after maintenance or repair work.

6.11 Low Voltage Switchgear and Control Gear

6.11.1 General requirement

- a) Switchgear assemblies shall be completely enclosed, self-supporting and suitable for floor or wall mounting, in multi-cubicle or multi-box type structures. The floor shall not be considered as being part of the enclosure.
- b) The assembly rated shall be suitable to form part of a system with coordinated insulation values in accordance with IEC 60664-1, overvoltage category IV, and shall be type tested accordingly.
- c) Partitions or barriers shall be provided in the assemblies to obtain Form 4 or 3B separation of compartments. Form 3B is only acceptable if all terminals have a degree of protection of at least IP 30. IEC 61439-2 defines forms of internal separation.
- d) The degree of protection as per IEC 60529 for any assembly shall be at least:
 - For indoor use (in enclosed buildings) IP 31
 - For outdoor use (protected by e.g. rain canopy) IP 55
- e) The requirements specified apply to type-tested assemblies (TTA). For assemblies used as sub-distribution systems, with incoming feeders protected by short-circuit current limiting devices of a nominal current of maximum 1600A at 480V.

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- f) Non-metallic enclosed switchgear and controlgear may be used for LV sub-distribution switchboards, when protected by short circuit current limiting devices having a maximum nominal current of 400 Amperes.
- g) Insulated plate shall be provided for cable entry to the panel
- h) Where generators are directly connected to an LV switchgear, the ratio of peak to rms short circuit current is likely to be higher than normal allowance, as per IEC standard. The peak short circuit requirement shall be clearly stated in the project specification.
- i) Unless otherwise specified in the requisition, the assembly shall be arranged for a TN-S power system, i.e. with separate neutral and protective busbars throughout the system.
- j) The main busbars shall be rated for the nominal current over the entire length of the assembly and shall not be rated lower than the incoming and sectionalizer units.
- k) A diversity factor of 1.0 shall be applied for temperature rise test or calculations for each switchboard panel containing only one functional unit. If a panel contains more than one two functional units, the rated diversity factor applied shall be not less than 0.9 when considering the circuit loads as known at the time of final inspection at the Manufacturer's works. Spare circuits and skeleton compartments shall be considered as being loaded to 90% of their rated current.
- l) All breakers and cables shall be loaded up to a maximum of 80% of its rated capacity.
- m) Fuse protection is preferred to moulded case circuit breakers (MCCB) with built-in protection. Additional protection relay shall be provided for incomer feeders rather than ACB/MCCB built-in protection.
- n) Switchgear nameplates including circuit labels, instruction and warning plates and caution notes shall be made of durable, corrosion resistant material.
- o) Nameplate information shall be mounted on the front of the assembly, e.g. on the sectionaliser or incoming feeder and include the following:
 - Name of Manufacturer,
 - Purchase order number,
 - Year of manufacture,
 - Type designation,
 - Rated voltage,
 - Rated frequency,
 - Rated current of busbar system,
 - Rated peak and short time withstand current and withstand time,
 - Degree of protection.
 - Other information to standards shall applied, include the relevant documents of the Manufacturer.

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- p) Warning plate(s) or caution notice(s) shall be installed in location to identifying the danger point(s).
- q) LED (Light-Emitting Diode) lamps shall be provided for indication/trip/warning purposes.
- r) Each circuit identification tag/number of the assembly shall be provided with circuit labels installed on the front of the fixed and the removable part of a withdrawal or removable unit.
- s) Assemblies shall be provided with a durable one-line synoptic diagram, indicating the busbar circuit and the specific function of the different compartments.
- t) The Manufacturer shall supply at least the following drawings/documents in quantities specify in the requisition:
 - Single-line diagram,
 - Schematic diagrams of all different types of circuits,
 - Documentation as specified in below for IMCS (Intelligent Motor Control Centre) only
 - General arrangement layout showing motor control unit/feeder control unit dimensions, central control unit dimensions, engineering workstation dimensions, mounting details of the units, connections, etc.
 - Cabling and wiring diagrams showing power input, output, terminals of motor/feeder control unit, central control unit, engineering workstation interface connections and include internal wiring cable schedule.
 - Setpoint setting tables for all units as set in the factory, hardcopy to be provided.
 - Configuration software for central/motor/feeder control unit.
 - Programming (logic) diagrams
 - Certificate of conformity for communication with FMCS/SCADA systems.
 - Arrangement drawing showing main circuits, main dimensions, panel layout, floor plan and shipping sections,
 - Minimum clearances during operation and maintenance around the assemblies for ventilation and safety,
 - Total mass of the assembly and of the individual shipping sections,
 - Transport, installation, commissioning, operation and maintenance instructions, limited and specific to the assembly and its components,
 - A list of recommended spare parts/components with model number,
 - Catalogs of protection relays, Power monitors, VFDs and other major components.
 - List of applicable type test certificates,
 - Test report of the final routine testing.

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- u) Some of the above documents may be combined, provided all information indicated above is included.
- v) All documents shall show the relevant order number, item and Manufacturer's references, and shall be distributed as specified in the purchasing documents.
- w) Assemblies supplied in transport units shall have these units clearly marked to facilitate assembly at site.
- x) Erection instructions and any special tools required for erection and maintenance shall form an integral part of the order and shall be packed with the equipment.
- y) The language used shall be as specified in the requisition. Danger and warning labels shall also be provided.
- z) All structural work shall be adequately protected against corrosion. Frame and partitions may be of galvanised steel without a further paint coating. Those parts/covers requiring painting shall be properly pre-treated before the final coat(s) of oil resistant finishing paint is (are) applied.
- za) Terminals shall be suitable for copper conductors.
- zb) Adequate terminals and cable termination space shall be provided for the main cabling due to the substantial cable derating. Phase separation barriers shall be provided at terminal block.
- zc) All switchgear components requiring maintenance shall be easily accessible. All components in wall or back-to-back mounted switchgear shall be accessible from the front only.
- zd) Exposed parts within the assemblies which have to be accessible during normal operation, maintenance or equipping of spare compartments, shall either not be live in the opened position or shall be protected to a degree of protection of at least IP 20.
- ze) Terminals of equipment installed on a compartment door and which can be live when the door is opened, shall be shrouded to a degree of protection of at least IP 20.
- zf) Compartment doors or covers shall be interlocked to prevent opening when the isolating switch/Circuit Breaker is in the 'ON' position. The isolating device/Circuit Breaker shall only be operable when the door is closed and when the withdrawable unit is in the inserted position.
- zg) All breakers in compartments shall be lockable in the OFF position to enable Lockout / tagout or CoHE.
- zh) Compartment doors and metal frame shall be electrically bonded to earth/ground point.
- zi) In the event of an internal fault in a functional unit, the damage shall be confined to that unit, so that the busbar system and all other functional units remain fit for further service. Likewise, an internal fault in a dropper system shall

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be confined to the associated panel, so that the busbar system and the other panels of the assembly remain fit for further service.

- zj) Assemblies shall comply with IEC 1641 with regard to internal arc testing.
- zk) Busbar system (phase, neutral and earth/ground) shall be of hard or medium hard drawn, high conductivity copper. Busbar should preferable be located on top of the switchboard assembly.

6.11.2 LV Distribution Board (Fixed type)

- a) All distribution switchboards shall comply with IEC 61439-4.
- b) Main LV distribution switchboards supplied by one or more generating sets shall have an incoming switch for each generator. The outgoing circuits shall be protected by fuses or ACB/MCCBs having a maximum rating/setting of 2500 A.
- c) Each sub-distribution switchboard shall be provided with one incoming switch.
- d) Each outgoing feeder shall be provided with a switching device with short circuit and residual current protection. The maximum outgoing feeder rating shall be 125A.
- e) The sensitivity of the residual current circuit breakers shall be as follows:
 - 30 mA for circuits below 125 A;
 - 300 mA for circuits of 125 A.
- f) Fuses shall be accessible only if the fuse bases are protected to at least IP 2X, or if interlocking exists so that the fuse base is electrically isolated first.
- g) Switchboard with redundant power supply which allows for paralleling, shall be rated for maximum short circuit capacity for both sources.
- h) Plain cable feeders for LV sub-distribution shall be controlled and protected by fuse-switch combinations or by MCCBs incorporating short circuit and earth/ground fault protective devices.
- i) ACB/MCCBs and fuses shall coordinate with outgoing circuit protective devices on the receiving-end sub-distribution switchboard.
- j) The rating of any fuse-protected LV circuit, which derives a supply from a distribution switchboard, should not exceed 50% of the rating of a fuse-protected LV circuit, which energises the switchboard. This is to ensure that the circuit protective devices can be fast operating and selective in isolating short circuits.

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6.11.3 LV Switchgear (Withdrawal type)

When a withdrawable unit has been removed from the assembly, the live parts inside the fixed compartment shall be protected against touch, with a degree of protection of at least IP 20. Where shutters are fitted to comply with above requirement, they shall be mechanically operated by the movement of the withdrawable unit and not be dependent on gravity

6.11.4 Circuit Breaker, Protective relay and other accessories

- 6.11.4.1 Circuit breakers in outgoing circuits shall comply with the requirements on operating accessibility, padlocking and phase and neutral switching.
- 6.11.4.2 Circuit breakers shall comply with and be type tested to IEC 947-2 and shall have utilisation category A, except for circuit breakers feeding sub-distribution switchboards which shall have utilisation category B.
- 6.11.4.3 Circuit breakers shall have a rated short-circuit making and breaking capacity in conformity with the prospective short-circuit current as specified in the requisition. Protective functions, such as short circuit, overload and earth/ground-fault, should be part of built-in features of the circuit breakers.
- 6.11.4.4 Circuit breakers in a change-over supply duty should be withdrawable but shall at least have isolating facilities for maintenance and testing purposes.
- 6.11.4.5 Depending on the circuit/system it protects, protective relays shall be intelligent electronic device (IED) type with built-in communication module based on IEC 61850.
- 6.11.4.6 Protective relays selected shall be able to operate continuously with internal supervision and parts should not be obsolete in at least the first 10 years of service. Manufacturer selected should ensure backward compatibility.
- 6.11.4.7 The following list of protective relays may be applicable:
 - i. Undervoltage relay (27)
 - ii. Overcurrent with instantaneous element (50/51), earth/ground-fault (50N/51N)
 - iii. Synchronizing relay (25)
 - iv. IED for motor protection (overload protection, earth/ground fault, stalled, etc.,)
 - v. Motor re-acceleration/restarting features (IED)
 - vi. Transformer differential (87T)
 - vii. Restricted/stand-by earth/ground fault

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6.11.4.8 Motor earth/ground fault protection for 30kW and above, if it is not incorporated in IED shall of core balance type. Relay shall be manually resettable after opening compartment. The sensitivity of the relays shall not exceed 3% of the rated nominal motor current or 6A, whichever is greater.

6.11.4.9 Switches/Isolators:

Switches/Isolators shall be easily accessible and operable from the front panel of the LV compartments without having to open the door.

Switches/Isolators shall be manual operating air-break with quick snap action in making and breaking the circuits.

Switches/Isolators shall comply with and be type tested to IEC 60947-3. AC 22 utilization category shall be used when switching mixed resistive and inductive loads. Utilization category AC 23 shall be for highly inductive loads.

Switches/Isolators shall be of the 4-pole type for 3-phases plus neutral system and 2-pole for single phase circuits. They shall be padlockable in the OFF position. Switches/Isolators shall be rated for short-circuit making capacity with prospective short-circuit current equal to required busbar system.

Switches/Isolators for incoming and bus coupler/section, if circuit breaker is not used, on a multi-section switchboard may have a short circuit rating based on actual distribution of the short circuit current. For example a coupled two-section board, about 50% of the maximum short circuit current will flow through each incomer.

Neutral pole of a switch/isolator in balanced 3 phase plus neutral system shall have current equal to other poles up to 63 amperes, inclusive. For devices above 63 A, neutral pole shall not be less than half of the current of other poles and a minimum rating of 63 A.

6.11.4.10 Metering:

- i. Each supply incoming unit shall be provided with one voltmeter connected to the busbar side between L1 and L3 phase. One ammeter measuring the L2 phase.

Metering supply to volt and watt-hour meters (if applicable) shall be protected with High breaking capacity (HBC-type) fuses of fully insulated pattern fixed directly onto the busbar system. A second set of fuses or miniature circuit breaker (MCB) with discrimination to the HRC fuses, shall be mounted locally near the instrument/s.

6.11.5 CT and VT

- a) Current transformers shall be in accordance with IEC 60185.
- b) Current transformers for measuring purposes shall be of accuracy class 3 (accuracy class 1 for measuring supplies from or to third parties) with an instrument security factor such that the instruments are not endangered at maximum fault current.
- c) The secondary current rating shall be 5 A or 1 A for ammeters and watt-hour meters mounted on the switchboard and 1 A for remote ammeters, either directly or via intermediate current transformers.
- d) Preferred primary ratings of instrument current transformers for remote ammeters are as follows:
10A, 20A, 30A, 50A, 75A, 100A, 150A, 200A, 300A, 400A, 500^a
- e) Protective current transformers shall be of accuracy class 5P except in cases where class 10P is specified as adequate by the relay Manufacturer.
- f) The short-circuit rating shall be at least 1 second if applied for short-circuit protection.
- g) The wiring of the secondary circuits shall have a cross section of at least 2.5 mm².
- h) Secondary circuits of all current transformers shall be earthed/grounded.
- i) Voltage Transformer (VT), if required, shall be in accordance with IEC 60186 and shall be of class 3 (Class 1 for measuring supplies from or to third parties).
- j) Secondary side of the VT of metering transformers shall be 110V.
- k) The secondary circuits of the voltage transformers shall be earthed/grounded.

6.11.6 Control Power

- a) Control supply for circuit breaker tripping and protective functions/relays (including IED) shall be taken from the substation DC UPS (refer DC UPS).
- b) Two separate DC power supplies with battery back-up shall be wired to a selector switch at the LV Switchgear. Selector switch shall be make-before-break and each DC supply source shall be connected via a blocking diode.
- c) Loss of DC supply at the LV switchgear shall be monitored and alarm to FMCS (& SCADA) system.

6.11.7 Automatic and Manual transfer operation

- a) LV switchgear shall be equipped control logic that allow both automatic transfer system (ATS) and manual transfer system (MTS).
- b) Provide Selector switch to choose between ATS or MTS at Tie Cubicle.
- c) Conditions taken into consideration when design ATS and MTS for LV Switchgear.
 - If LV switchgear bus coupler (Tie) is always in closed position (NC), then ATS function is not required. MTS is still required.
 - If LV switchgear bus coupler (Tie) is always in open position (NO), then LV switchgear require both function ATS and MTS.
- d) LV switchgear shall be able to perform Auto-Transfer and Auto-Restore during testing and troubleshooting when ATS function is available.
- e) Automatic transfer system operation shall meet all criterion prior to initiation.
 - Both Incoming circuit breakers (NC), tie circuit breaker (NO), are in normal position, in-service, healthy.
 - Permanent Power outage occurs at one side of Incoming line. This must be confirmed by at least two detections of undervoltage relays (27) at different point of measurement (Incoming line voltage and Bus voltage). Another Incoming line must be healthy, normal voltage, and free from bus fault or any fault.
 - After confirming item above, dead side Incoming line shall be isolated and opened. Associated dead bus must be health and free from any bus fault or any fault.
- f) After items are met then load transfer can be initiated by closing tie circuit breaker.

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- g) Manual Restoration Operation should be recommended during normal power recovery.
- h) Manual Restoration/Transfer Operation
 - System is normal with no protective function activated.
 - Outage supply line is healthy and ready to take load with line undervoltage relay (27) normalized.

6.11.8 Installation

6.11.8.1 General Requirements

- i. Switchgear shall be installed in accordance with the installation instructions and supporting drawing provided by the Manufacturer. This include switchboard foundation and any inserts to be cast and within specific tolerances.
- ii. Switchboard shall be installed, as defined by Manufacturer, with sufficient space to facilitate maintenance.
- iii. Foundation drawings, together with information on mass and loading data, shall be provided to permit correct civil design of building and foundations.
- iv. If switchboards and/or controlgear are from different Manufacturer, one composite foundation drawing shall be supplied.
- v. Equipment arrangement and layout drawing shall be provided by Manufacturer or supplier for it to be incorporated into the overall arrangement drawings and cabling design.
- vi. Substation floors shall be smooth and level to permit the handling of equipment on rollers. Concrete floors shall have a surface protection layer to avoid formation of dust.
- vii. Switchboard shall only be installed after the completion of switchroom civil and building works, this is to minimise the ingress of dust and dirt during or after the erection.
- viii. Any modifications required during installation shall be approved by Owner prior to implementation. All modifications shall be recorded and information incorporated into the "as-built" drawings.
- ix. Switchboards shall be installed in location that minimise the risk of vibration at vicinity. If unavoidable, anti-vibration mountings may be used.

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- x. Switchboard installed shall avoid any pipes work in vicinity. If unavoidable, pipes should not have joints at that area.
- xi. All cables shall be enter from the bottom or side of the switchboard, for cases where the installation is outside of the electrical room, unless the switchboard is installed in electrical or control room with properly insulated and covered, to avoid potential risk caused by water, liquid...etc.

6.12 Low Voltage Motor Control Centre (LV MCC)

6.12.1 General

- a) LV MCC shall be design and built to comply to IEC 61439 or UL845 or NEMA ICS 18 standard include arc resistance construction and arcflash containment.
- b) Only natural cooling shall be applied for LV MCC
- c) Refer to LV Switchgear and controlgear general requirement (6.11.1) shall applied.
- d) A space of at least 0.2 metre shall be available from the floor to any measuring apparatus, switch, indication light, push button or any other components. Maximum height of the above components shall not be more than 2 metres above the floor level. Also refer to NFPA 17 or UL standard 845 for compliances.
- e) Withdrawal units which is electrical and mechanical identical shall be interchangeable.

6.12.2 Switches/Isolators:

- a) Switches/Isolators shall be easily accessible and operable from the front panel of the LV MCC compartments without having to open the compartment cover.
- b) Switches/Isolators shall be manual operating air-break with quick snap action in making and breaking the circuits.
- c) Switches/Isolators shall comply with and be type tested to IEC 60947-3. AC23 utilization category shall be used for motor switching loads.
- d) Switches/Isolators shall be of the 4-pole type for 3-phases plus neutral system and 2-pole for single phase circuits. They shall be padlockable in the OFF position.
- e) Switches/Isolators shall be rated for short-circuit making capacity with prospective short-circuit current equal to required busbar system.
- f) Switches/Isolators for incoming and bus coupler/section, if circuit breaker is not used, on a multi-section switchboard may have a short circuit rating based on actual distribution of the